

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 3

HINTS

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We will discuss several questions from Chapter 4 in Ross' book.

Tutorial Questions

- (1) Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:
- (a) the maximum value to appear in the two rolls;
 - (b) the minimum value to appear in the two rolls;
 - (c) the sum of the two rolls;
 - (d) the value of the first roll minus the value of the second roll?

Solution:

- (a) 1, 2, 3, 4, 5, 6;
 - (b) 1, 2, 3, 4, 5, 6;
 - (c) 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12;
 - (d) -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5.
- (2) A salesman has scheduled two appointments to sell encyclopedias. His first appointment will lead to a sale with probability .3, and his second will lead independently to a sale with probability .6. Any sale made is equally likely to be either for the deluxe model, which costs \$1000, or the standard model, which costs \$500. Determine the probability mass function of X , the total dollar value of all sales.

Solution: The range of X is $\{0, 500, 1000, 1500, 2000\}$. We use the following abbreviations to denote events:

F : the first appointment leads to a sale;
 S : the second appointment leads to a sale;
 F_1 : getting a sale of 500 at the first appointment;
 F_2 : getting a sale of 1000 at the first appointment;
 S_1 : getting a sale of 500 at the second appointment;
 S_2 : getting a sale of 1000 at the second appointment.

Then

$$F_1, F_2 \subseteq F, \quad S_1, S_2 \subseteq S$$

and

$$P(F') = 1 - P(F) = 1 - 0.3 = 0.7, \quad P(S') = 1 - P(S) = 1 - 0.6 = 0.4,$$

and

$$P(F_1|F) = P(F_2|F) = P(S_1|S) = P(S_2|S) = 0.5.$$

For $i = 1, 2$, we have, by the law of total probability:

$$P(F_i) = P(F_i|F)P(F) + P(F_i|F')P(F') = (0.5)(0.3) + 0(0.7) = 0.15,$$

and

$$P(S_i) = P(S_i|S)P(S) + P(S_i|S')P(S') = (0.5)(0.6) + 0(0.4) = 0.3.$$

Since two appointments are independent of each other,

$$P(X = 0) = P(F' \text{ and } S') = P(F')P(S') = (1 - 0.3)(1 - 0.6) = 0.28;$$

$$\begin{aligned} P(X = 500) &= P((F_1 \text{ and } S') \text{ or } (F' \text{ and } S_1)) \\ &= P(F_1)P(S') + P(F')P(S_1) = (0.15)(0.4) + (0.7)(0.3) = 0.27; \end{aligned}$$

$$\begin{aligned} P(X = 1000) &= P((F_2 \text{ and } S') \text{ or } (F_1 \text{ and } S_1) \text{ or } (F' \text{ and } S_2)) \\ &= P(F_2)P(S') + P(F_1)P(S_1) + P(F')P(S_2) = (0.15)(0.4) + (0.15)(0.3) + (0.7)(0.3) = 0.315; \end{aligned}$$

$$\begin{aligned} P(X = 1500) &= P((F_1 \text{ and } S_2) \text{ or } (F_2 \text{ and } S_1)) \\ &= P(F_1)P(S_2) + P(F_2)P(S_1) = 2(0.15)(0.3) = 0.09 \end{aligned}$$

and

$$P(X = 2000) = P(F_2 \text{ and } S_2) = P(F_2)P(S_2) = (0.15)(0.3) = 0.045.$$

(3) Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0 \\ \frac{b}{4}, & 0 \leq b < 1 \\ \frac{1}{2} + \frac{b-1}{4}, & 1 \leq b < 2 \\ \frac{11}{12}, & 2 \leq b < 3 \\ 1, & 3 \leq b \end{cases}$$

(a) Find $P\{X = i\}$, $i = 1, 2, 3$.

(b) Find $P\{\frac{1}{2} < X < \frac{3}{2}\}$.

Solution:

(a)

$$\begin{aligned} P\{X = 1\} &= F(1) - F(1^-) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \\ P\{X = 2\} &= F(2) - F(2^-) = \frac{11}{12} - \frac{3}{4} = \frac{1}{6} \\ P\{X = 3\} &= F(3) - F(3^-) = 1 - \frac{11}{12} = \frac{1}{12} \end{aligned}$$

(b)

$$P\left\{\frac{1}{2} < X < \frac{3}{2}\right\} = P\left\{X < \frac{3}{2}\right\} - P\left\{X \leq \frac{1}{2}\right\} = F\left(\frac{3}{2}\right) - F\left(\frac{1}{2}\right) = \frac{5}{8} - \frac{1}{8} = \frac{1}{2}.$$

(4) Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by team A with probability p . Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show that in both cases that this number is maximized when $p = 1/2$.

Solution:

(a) Assume that $i = 2$. Let X be the number of games played and X can only take value 2 or 3. It is easy to see that $P\{X = 2\} = p^2 + (1-p)^2$ and $P\{X = 3\} = 2p(1-p)$. Hence

$$\mathbb{E}(X) = (p^2 + (1-p)^2) \cdot 2 + 2p(1-p) \cdot 3 = 2(1 + p - p^2).$$

Note that $p - p^2$ is maximized when $p = 1/2$, it follows that $\mathbb{E}(X)$ is maximized when $p = 1/2$.

- (b) Assume that $i = 3$. Let X be the number of games played and X can only take value in $\{3, 4, 5\}$. It is easy to see that

$$\begin{aligned} P\{X = 3\} &= p^3 + (1 - p)^3, \\ P\{X = 4\} &= 3p^3(1 - p) + 3p(1 - p)^3, \\ P\{X = 5\} &= 6p^3(1 - p)^2 + 6p^2(1 - p)^3 = 6p^2(1 - p)^2. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}(X) &= 3(p^3 + (1 - p)^3) + 4(3p^3(1 - p) + 3p(1 - p)^3) + 5 \cdot 6p^2(1 - p)^2 \\ &= 3(1 + p + p^2 - 4p^3 + 2p^4). \end{aligned}$$

Then

$$\frac{d}{dp} \mathbb{E}(X) = 3(1 + 2p - 12p^2 + 8p^3) = 3(1 - 2p)(1 + 4p - 4p^2),$$

which is 0 when $p = 1/2$, and is positive when $p < 1/2$ and negative when $p > 1/2$. Hence the maximum is attained when $p = 1/2$.

- (5) One of the numbers 1 through 10 is randomly chosen. You are to try to guess the number chosen by asking questions with yes-no answers. Compute the expected number of questions you will need to ask in each of the following two cases:
- Your i -th question is to be “Is it i ?”, $i = 1, 2, \dots, 10$.
 - With each question you try to eliminate one half of the remaining numbers, as nearly as possible.

Solution:

- (a) If the true number is i you need to guess i times. Hence the expected number of questions is

$$\sum_{i=1}^{10} \frac{i}{10} = \frac{55}{10} = \frac{11}{2}.$$

- (b) Let N_i be the number of possible answers left after the i -th question. Initially $N_0 = 10$. The first question will reduce the number of candidates down to $N_1 = 5$. (For instance, you can ask “Is the number at least 5.5?” or “Is the number odd?”.) For the second question, you can reduce the number of candidates down to $N_2 \in \{2, 3\}$. Furthermore,

$$P\{N_2 = 2\} = \frac{2}{5}, \quad P\{N_2 = 3\} = \frac{3}{5}.$$

Next we need to do a bit case analysis.

- When $N_2 = 2$, only one more question is needed and thus $X = 3$.
- When $N_2 = 3$, we ask one more question and obtain $N_3 \in \{1, 2\}$, where

$$P\{N_3 = 1|N_2 = 3\} = \frac{1}{3}, \quad P\{N_3 = 2|N_2 = 3\} = \frac{2}{3}.$$

- When $N_3 = 1$, we know it is the answer and thus $X = 3$.
- When $N_3 = 2$, we need to ask one more question to find out the answer, and thus $X = 4$.

To summarize,

$$P\{X = 3\} = P\{N_2 = 2\} + P\{N_2 = 3, N_3 = 1\} = \frac{2}{5} + \frac{3}{5} \cdot \frac{1}{3} = \frac{3}{5}$$

$$P\{X = 4\} = P\{N_2 = 3, N_3 = 2\} = \frac{3}{5} \cdot \frac{2}{3} = \frac{2}{5}.$$

Therefore

$$\mathbb{E}(X) = 3 \cdot \frac{3}{5} + 4 \cdot \frac{2}{5} = \frac{17}{5}.$$

- (6) If $\mathbb{E}(X) = 1$ and $\text{Var}(X) = 5$, find

(a) $\mathbb{E}((2 + X)^2)$;

(b) $\text{Var}(4 + 3X)$.

(We assume that X is a discrete random variable.)

Solution:

- (a) $\mathbb{E}((2 + X)^2) = \mathbb{E}(4 + 4X + X^2) = 4 + 4\mathbb{E}(X) + \mathbb{E}(X^2)$. Since $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$, we can solve that $\mathbb{E}(X^2) = 6$. Hence

$$\mathbb{E}((2 + X)^2) = 4 + 4 \cdot 1 + 6 = 14.$$

- (b) $\text{Var}(4 + 3X) = 9\text{Var}(X) = 45$.

- (7) Suppose that, in flight, airplane engines will fail with probability $1 - p$, independently from engine to engine. If an airplane needs a majority of its engines operative to complete a successful flight, for what values of p is a 5-engine plane preferable to a 3-engine plane?

Solution: We want the probability that at most 2 of 5 engines fail is bigger than the probability that at most 1 of 3 engines fail.

The probability that at most 2 of 5 engines fail is

$$p^5 + 5(1 - p)p^4 + 10(1 - p)^2p^3$$

and the probability that at most 1 of 3 engines fail is

$$p^3 + 3(1 - p)p^2.$$

So we want

$$p^5 + 5(1 - p)p^4 + 10(1 - p)^2p^3 > p^3 + 3(1 - p)p^2.$$

Divide p^2 on both sides,

$$p^3 + 5(1 - p)p^2 + 10(1 - p)^2p > p + 3(1 - p),$$

rearranging the terms,

$$3(1 - p)^2(1 - 2p) < 0$$

and thus

$$p > \frac{1}{2}.$$

- (8) It is known that diskettes produced by a certain company will be defective with probability 0.01, independently of each other. The company sells the diskettes in packages of size 10 and offers a money-back guarantee that at most 1 of the 10 diskettes in the package will be defective. The guarantee is that the customer can return the entire package of diskettes if he or she finds more than one defective diskette in it. If someone buys 3 packages, what is the probability that he or she will return exactly 1 of them?

Solution: Within each package, the probability that there are more than 1 defective items is

$$1 - \binom{10}{0}0.01^00.99^{10} - \binom{10}{1}0.01^10.99^9 \approx 0.0043.$$

Hence the probability that exactly 1 of 3 packages is bad is

$$\binom{3}{1}0.0043^1 \cdot (1 - 0.0043)^2 \approx 0.0128.$$

Assignment Questions

- (1) Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X ? For $n = 3$, find the probabilities associated with the values that X can take on? (We assume the coin is assumed fair.)

Solution: Let t be the number of tails obtained. Then $X = |(n - t) - t| = |n - 2t|$, where $t \in \{0, 1, \dots, n\}$. The values X can take depends on whether n is odd or even:

- If n is even, the values X can take are $n, n - 2, \dots, 2, 0$.
- If n is odd, the values X can take are $n, n - 2, \dots, 1$.

For $n = 3$, the values X can take are 3, 1, and

$$P\{X = 3\} = P(H = 3 \text{ \& } T = 0) + P(H = 0 \text{ \& } T = 3) = 2 \cdot \binom{3}{0} (0.5)^3 = 0.25,$$

$$P\{X = 1\} = P(H = 2 \text{ \& } T = 1) + P(H = 1 \text{ \& } T = 2) = 2 \cdot \binom{3}{2} (0.5)^3 = 0.75.$$

- (2) In the game of Two-Finger Morra, 2 players show 1 or 2 fingers and simultaneously guess the number of fingers their opponent will show. If only one of the players guesses correctly, he wins an amount by him and his opponent. If both players guess correctly or if neither guesses correctly, then no money is exchanged. Consider a specified player, and denote by X the amount of money he wins in a single game of Two-Finger Morra.
- (a) If each player acts independently of the other, and if each player makes his choice of the number of fingers he will hold up and the number he will guess that his opponent will hold up in such a way that each of the 4 possibilities is equally likely, what are the possible values of X and what are their associated probabilities?
- (b) Suppose that each player acts independently of the other. If each player decides to hold up the same number of fingers that he guesses his opponent will hold up, and if each player is equally likely to hold up 1 or 2 fingers, what are the possible values of X and their associated probabilities?

Solution: (a) The possible values of X are 0, if both or neither player guesses correctly, 2, 3, or 4, if he guesses correctly and the other player doesn't, and -2 , -3 , or -4 , if the other player guesses correctly and he doesn't.

Thus,

$$P\{X = 2\} = P(\text{he guesses 1}) \cdot P(\text{he holds up 1}) \cdot P(\text{the other player guesses 2})$$

$$\cdot P(\text{the other player holds up 1}) = \frac{1}{24} = \frac{1}{16}.$$

The same argument gives

$$P\{X = -2\} = P\{X = 4\} = P\{X = -4\} = \frac{1}{16},$$

and

$$P\{X = -3\} = P\{X = 3\} = \frac{1}{8}.$$

This gives

$$P\{X \neq 0\} = \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2},$$

and hence

$$P\{X = 0\} = \frac{1}{2}.$$

- (b) If both players guess the same, then both players guess correctly and $X = 0$. If both players guess differently, then both players are wrong and $X = 0$. Thus the only possible value for X is 0 and hence $P\{X = 0\} = 1$.
- (3) Five distinct numbers are randomly distributed to players numbered 1 through 5. Whenever two players compare their numbers, the one with the higher one is declared the winner. Initially, players 1 and 2 compare their numbers; the winner then compares her number with that of player 3, and so on. Let X denote the number of times player 1 is a winner. Find $P\{X = i\}$, $i = 0, 1, 2, 3, 4$.

Solution: Note the first and the second player have equal probability to get any of number. This gives

$$P\{X = 0\} = 1/2.$$

Event $X = 1$ means that the first player has got greater number than the second player, but not than the third player. So, choose any three numbers out of the five of them and say that the minimal number out of these three goes to the second player, the mean number to the first one and the largest number to the third one. Permute remaining two numbers on remaining two people. This gives

$$P\{X = 1\} = \frac{\binom{5}{3} \cdot 2!}{5!} = 1/6.$$

Even $X = 2$ means that the first player has got greater number than the second and the third player, but not than the fourth player. So choose any 4 numbers out of five of them and say that the minimal number and the next minimal out of these four go to the second and third player (and permute them), third number to the first one, and the largest to the fourth player. Give the remaining to the last person. We have

$$P\{X = 2\} = \frac{\binom{5}{4} \cdot 2!}{5!} = 1/12.$$

Even $X = 3$ means that the first player has got greater number than the second, the third and the fourth player, but not than the fifth player. So permute these five numbers as follows: give the highest to the last person, the second highest to the first, and permute remaining numbers on the remaining people. We have

$$P\{X = 3\} = \frac{3!}{5!} = 1/20.$$

Even $X = 4$ means basically that the first player has won all the battles, i.e. he has got the greatest number. Thus

$$P\{X = 4\} = \frac{4!}{5!} = 1/5.$$

- (4) A gambling book recommends the following “winning strategy” for the game of roulette: Bet \$1 on red. If red appears (which has probability $\frac{18}{38}$), then take the \$1 profit and quit. If red does not appear and you lose this bet (which has probability $\frac{20}{38}$ of occurring), make additional \$1 bets on red on each of the next two spins of the roulette wheel and then quit. Let X denote your winnings when you quit.
- (a) Find $P\{X > 0\}$.
- (b) Are you convinced that the strategy is indeed a “winning” strategy? Explain your answer.
- (c) Find $\mathbb{E}(X)$.

Solution: (a) Note the possible outcomes of X are 1, -1 , -3 . The gambler will win 1\$ if he wins in the first bet or he loses the first bet, but he has success in the second and the third. He will win -1 \$ if he loses the first and the second, but wins the third, or

loses the first and the third, but wins the second. Finally, he will win $-3\$$, if he loses all three bets. Thus,

$$P\{X > 0\} = P\{X = 1\} = \frac{18}{38} + \frac{20}{38} \left(\frac{18}{38}\right)^2 = \frac{4059}{6859} \approx 0.5918.$$

(b) No. This is not the winning strategy, even though the probability that the gambler wins is bigger than $1/2$.

(c) Consider

$$P\{X = -1\} = 2 \left(\frac{20}{38}\right)^2 \frac{18}{38} = \frac{1800}{6859},$$

$$P\{X = -3\} = \left(\frac{20}{38}\right)^3 = \frac{1000}{6859},$$

and we have the expectation

$$\mathbb{E}(X) = 1 \times \frac{4059}{6859} + (-1) \times \frac{1800}{6859} + (-3) \times \frac{1000}{6859} \approx -0.11$$

which says that the gambler is expected to lose 11 cents.

- (5) Four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively, 40, 33, 25, and 50 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on her bus.

- (a) Which of $\mathbb{E}(X)$ or $\mathbb{E}(Y)$ do you think is larger? Why?
 (b) Compute $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.

Solution: (a) $\mathbb{E}(X)$ is larger because a student is more likely to be chosen from a bus with larger number of students, but for $\mathbb{E}(Y)$ the bus driver is selected equally.

(b) The table below shows the probability for X :

x	40	33	25	50
$P\{X = x\}$	0.270	0.223	0.169	0.338

Thus,

$$\mathbb{E}(X) = 40 \times 0.270 + 33 \times 0.223 + 25 \times 0.169 + 50 \times 0.338 = 39.28.$$

$\mathbb{E}(Y)$ is based on the probability that “1 of the four bus drivers is randomly selected”, $P(X = x) = 1/4$. Thus,

$$\mathbb{E}(Y) = 40 \times 0.25 + 33 \times 0.25 + 25 \times 0.25 + 50 \times 0.25 = 37.$$

- (6) Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by team A with probability p . Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show in both cases that this number is maximized when $p = 1/2$.

Solution: Define X as the random variable that marks the number of games that are played. Observe that there are maximally played 3 games and minimally 2 games. Hence $X \in \{2, 3\}$.

For $X = 2$, some of the team has won both of the first two games, and hence

$$P\{X = 2\} = p^2 + (1 - p)^2.$$

For $X = 3$, some of the team need won two games and the other team has to win one game of the first two. Thus

$$P\{X = 3\} = 2p(1 - p)^2 + 2p^2(1 - p) = 2p(1 - p).$$

This gives

$$\mathbb{E}(X) = 2 \cdot (p^2 + (1-p)^2) + 3 \cdot 2p(1-p) = -2p^2 + 2p + 2,$$

which takes maximum value when $p = 1/2$.

(b) Assume that $i = 3$. Let X be the number of games played and X can only take value in $\{3, 4, 5\}$. It is easy to see that

$$\begin{aligned} P\{X = 3\} &= p^3 + (1-p)^3, \\ P\{X = 4\} &= 3p^3(1-p) + 3p(1-p)^3, \\ P\{X = 5\} &= 6p^2(1-p)^2. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}(X) &= 3(p^3 + (1-p)^3) + 4(3p^3(1-p) + 3p(1-p)^3) + 5 \cdot 6p^2(1-p)^2 \\ &= 3(1 + p + p^2 - 4p^3 + 2p^4). \end{aligned}$$

Then

$$\frac{d}{dp} \mathbb{E}(X) = 3(1 + 2p - 12p^2 + 8p^3) = 3(1 - 2p)(1 + 4p - 4p^2),$$

which is 0 when $p = 1/2$, and is positive when $p < 1/2$ and negative when $p > 1/2$. Hence the maximum is attained when $p = 1/2$.

- (7) You have \$1000, and a certain commodity presently sells for \$2 per ounce. Suppose that after one week the commodity will sell for either \$1 or \$4 an ounce, with these two possibilities being equally likely.
- (a) If your objective is to maximize the expected amount of money that you possess at the end of the week, what strategy should you employ?
 - (b) If your objective is to maximize the expected amount of the commodity that you possess at the end of the week, what strategy should you employ?

Solution: (a) Let X be the amount of money you possess at the end of the week. Let k be the number of items you buy to begin with. After a week, in order to maximize X , you must sell all k items. We are given

$$P\{X = 1000 - 2k + k\} = P\{X = 1000 - 2k + 4k\} = 1/2,$$

and hence

$$\mathbb{E}(X) = (1000 - 2k + k)\frac{1}{2} + (1000 - 2k + 4k)\frac{1}{2} = 1000 + k/2.$$

To maximize it, we need k to be as large as possible, and hence, we use \$1000 to buy 500 items initially, and then sell them all after one week.

(b) Let Y be the amount of commodity that you possesses at the end of the week. Let k be the number of items you buy to begin with. After a week, to maximize Y , you must buy as many items as possible. If the price at the end of the week is \$1, then you will be able to buy $1000 - 2k$ items. If the price at the end of the week is \$4, then you will be able to buy $\left\lfloor \frac{1000-2k}{4} \right\rfloor$ items, where $\left\lfloor \frac{1000-2k}{4} \right\rfloor$ is the largest integer not bigger than $\frac{1000-2k}{4}$. Thus,

$$\begin{aligned} \mathbb{E}(Y) &= (1000 - 2k + k)\frac{1}{2} + \left(\left\lfloor \frac{1000 - 2k}{4} \right\rfloor + k \right) \frac{1}{2} \\ &= \left(1000 + \left\lfloor \frac{1000 - 2k}{4} \right\rfloor \right) \frac{1}{2}. \end{aligned}$$

To maximize this, we want k as small as possible. That is, the best strategy is to buy no items initially and to buy as many items as possible one week later.

- (8) Each night different meteorologists give us the probability that it will rain the next day. To judge how well these people predict, we will score each of them as follows: If a meteorologist says that it will rain with probability p , then he or she will receive a score of

$$\begin{cases} 1 - (1 - p)^2 & \text{if it does rain} \\ 1 - p^2 & \text{if it does not rain} \end{cases}$$

We will then keep track of scores over a certain time span and conclude that the meteorologist with the highest average score is the best predictor of weather. Suppose now that a given meteorologist is aware of our scoring mechanism and wants to maximize his or her expected score. If this person truly believes that it will rain tomorrow with probability p^* , what value of p should he or she assert so as to maximize the expected score?

Solution: Let X be the random variable that marks the score of a certain meteorologist at some day. If he says that it will win (he says it with probability p^*) and it really rains (with probability p), he gets score $1 - (1 - p)^2$. Also, if he says that it will not win (he says it with probability $1 - p^*$) and it really does not rain (with probability $1 - p$), he gets score $1 - p^2$. Thus

$$\mathbb{E}(X) = p^*(1 - (1 - p)^2) + (1 - p^*)(1 - p^2) = 2pp^* - p^* - p^2 + 1.$$

Analyse the function $p \mapsto \mathbb{E}(X)$ and find its maximum on $(0, 1)$ by

$$\frac{d}{dp} \mathbb{E}(X) = 2p^* - 2p = 0,$$

giving $p^* = p$, which means that the meteorologist maximizes its chance when $p^* = p$, coincide our intuition.